

Analyzing Graduation Admission Data Using Bayesian Beta Regression: A Probabilistic Approach

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Abstract

This study investigates the use of a Bayesian Beta Regression model for predicting academic admission probabilities, emphasizing its performance, convergence, and predictive accuracy. The model was evaluated using Markov Chain Monte Carlo (MCMC) methods, with convergence diagnostics such as Gelman-Rubin and Geweke indicating reliable results. Despite some autocorrelation in the chains, the model exhibited satisfactory mixing and independence. Various evaluation metrics, including the Watanabe-Akaike Information Criterion (WAIC), Leave-One-Out Cross-Validation (LOO-CV), and Mean Squared Error (MSE), were employed to assess the model's fit and prevent overfitting. Posterior predictive checks confirmed the model's high predictive accuracy, with low MSE and a favorable posterior predictive p-value (PPP). A comparative analysis with existing models demonstrated the superior predictive performance of the proposed approach. The findings suggest that the Bayesian Beta Regression model provides a robust and effective tool for decision-making in academic admission processes, offering valuable insights for future applications in complex decision-making scenarios.

Keywords: Bayesian Beta Regression, Markov Chain Monte Carlo (MCMC), Convergence Diagnostics, Watanabe-Akaike Information Criterion (WAIC), Leave-One-Out Cross-Validation (LOO-CV), and Mean Squared Error (MSE), Posterior Predictive Checks

INTRODUCTION

Education is a broad term encompassing knowledge, experience, learning, and teaching. It plays a crucial role in shaping individuals and societies, fostering personal and social development. It fosters personal and social development, helping individuals cultivate critical thinking, problem-solving, decision-making, and communication

skills, which are essential for adapting to and thriving in an ever-evolving world. As John Dewey stated, “Education is the process of living through a continuous reconstruction of experiences.”

In the 21st century, education is rapidly evolving to meet the demands of a knowledge-based economy. Driven by global interconnectedness and technological advancements, the expectations placed on learners have expanded beyond traditional academic achievements. To succeed in this dynamic landscape, individuals must acquire a diverse set of skills, including creativity, adaptability, collaboration, and digital literacy (Kist, 2003; Kennedy & Sundberg, 2020). These shifts highlight the growing need to understand the factors influencing student performance and how education systems can equip learners for the challenges of the modern world.

Despite the increasing emphasis on holistic education, many students face challenges in achieving academic success due to a combination of academic, personal, and environmental factors. While variables such as study habits, extracurricular activities, sleep patterns, and prior academic achievements are often acknowledged as influential, their specific relationships with academic performance remain underexplored. Using Bayesian Beta Regression, the research examines how predictors such as GRE and TOEFL scores, undergraduate GPA, research experience, and university ratings affect the Chance of Admit, a continuous variable bounded between 0 and 1. The findings aim to provide actionable insights for prospective students to optimize their applications and for admissions committees to refine evaluation processes, fostering a fairer and more effective system.

METHODOLOGY

Our dataset consists of data collected from 500 students, each with various factors that influence the likelihood of graduate admission. These factors include GRE scores, TOEFL scores, undergraduate GPA, research experience, and university ratings. The target variable for this study is the "Chance of Admit," a continuous variable representing the probability of a student being admitted to a graduate program.

When predicting values that are within a bound range linear prediction models usually end up predicting for more than what the limit is. It would be great if our chosen prediction model would account for our target value's bounded range. Unlike a normal distribution a beta distribution uses two unintuitive parameters to create a range

for our values, The parameters can be found using the mean and precision of our data just like a normal distribution.

Given that the "Chance of Admit" falls within the range of 0 and 1, it is crucial to select a model that can appropriately handle this bounded nature. The Bayesian beta regression is well-suited for modeling continuous variables that are constrained to a specific interval, such as the "Chance of Admit", it uses two parameters that are derived from the data's mean and precision, which are analogous to the mean and variance used in normal distributions. This enables the model to effectively capture variability within the bounded range.

The response for "Chance of Admit", was modeled using a beta distribution, with the mean linked to the predictors through a logit function and a precision parameter. Predictors such as GRE, TOEFL, and GPA were standardized to improve convergence. The model was implemented in JAGS, using weakly informative normal priors ($\text{Normal}(0, 10^{-6})$) for regression coefficients and a gamma prior ($\text{Gamma}(0.001, 0.001)$) for the precision parameter.

Parameter estimation was performed using Markov Chain Monte Carlo (MCMC) sampling via JAGS, with four chains of 2000 iterations each and a 500-iteration burn-in phase. Model diagnostics, including trace plots and the Gelman-Rubin statistic, confirmed convergence. Posterior predictive checks validated the model's fit, showing close agreement between observed and predicted values. Predictive accuracy was assessed using cross-validation and Mean Squared Error (MSE) metrics, ensuring reliable and interpretable parameter estimates.

RESULTS

This section presents the findings of Beta Regression analysis, which is focused on understanding the factors influencing Chance of Admit, a proportion-based response variable bounded between 0 and 1. The model is evaluated through sampling using JAGS to assess its general performance and find Chance of Admit's key predictors.

Iterations = 2001:7000
Thinning interval = 1
Number of chains = 5
Sample size per chain = 5000

1. Empirical mean and standard deviation for each variable,
plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
beta[1]	0.13120	0.03313	0.0002095	0.0006461
beta[2]	0.11360	0.03232	0.0002044	0.0006016
beta[3]	0.04489	0.02530	0.0001600	0.0003467
beta[4]	-0.01406	0.02583	0.0001634	0.0003828
beta[5]	0.08179	0.02249	0.0001423	0.0002871
beta[6]	0.37348	0.03354	0.0002121	0.0006425
beta[7]	0.04218	0.01956	0.0001237	0.0002122
beta_0	1.06103	0.01667	0.0001054	0.0001423
phi	50.80270	3.60696	0.0228124	0.0298421

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
beta[1]	0.065163	0.10908	0.13123	0.153650	0.19596
beta[2]	0.050445	0.09215	0.11355	0.135171	0.17699
beta[3]	-0.003968	0.02769	0.04485	0.061881	0.09463
beta[4]	-0.064201	-0.03138	-0.01405	0.003252	0.03622
beta[5]	0.038101	0.06649	0.08174	0.097036	0.12593
beta[6]	0.308166	0.35090	0.37305	0.395773	0.44039
beta[7]	0.004180	0.02909	0.04198	0.055347	0.08072
beta_0	1.028574	1.04978	1.06105	1.072149	1.09375
phi	44.011474	48.26381	50.71266	53.211603	57.99493

Table 1. Statistical Summary of the Samples

The analysis reveals that CGPA and GRE scores have the most positive effects on admission chances, with mean values of 0.37348 and 0.1312 and credible intervals that exclude zero, confirming that higher GRE and TOEFL scores are strongly associated with increased admission probabilities.

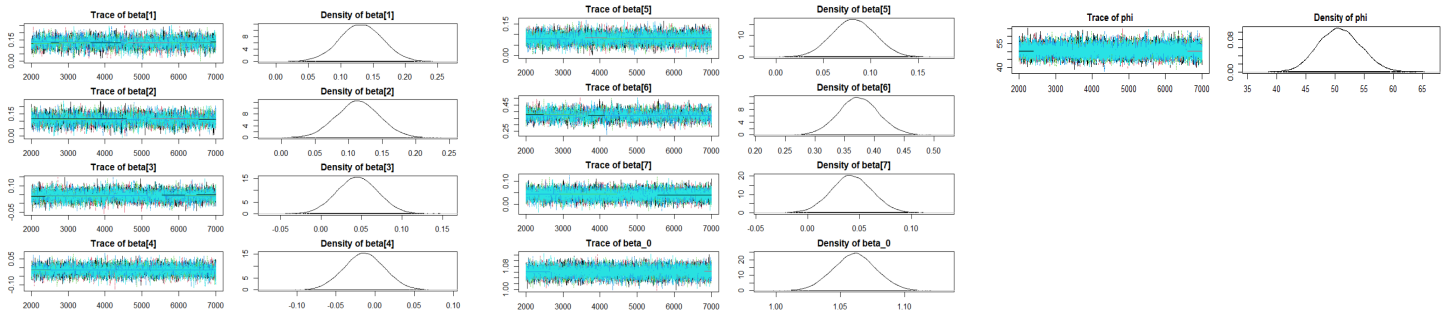


Figure 1. Trace & Density plot of beta1, beta2, beta3, beta4, beta5, beta6, beta7, beta0, and phi

According to the trace plot and density plot, the results show that the parameters, ranging from beta_0 to phi, have successfully converged. This convergence indicates that the model has stabilized and is providing reliable estimates, particularly for the key predictors of the chance of admission, confirming the consistency and accuracy of the model's predictions.

```

Chain 1
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7] beta_0 phi
-1.7378 1.6426 -2.4203 -0.4089 0.4455 0.8929 2.0861 -0.9663 -0.3063

Chain 2
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7] beta_0 phi
0.5615 -1.0057 0.9255 -1.4266 0.3220 0.5747 -0.5469 -1.3705 0.9675

Chain 3
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7] beta_0 phi
-0.70905 -0.78897 -0.53825 -0.05437 -1.01871 1.89991 0.56635 -0.84314 -1.25545

Chain 4
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7] beta_0 phi
0.7101 -0.5322 0.2100 0.6352 -0.7630 -0.4713 0.4204 0.4869 -0.2498

Chain 5
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7] beta_0 phi
0.4121 -0.5882 1.3565 -0.3680 0.3430 -0.2404 -0.5204 0.9498 1.0835

```

Table 2. Geweke values of chain 1 to chain 5

The Geweke values provided to us from running a Geweke Diagnostics check we can assume that in the first chain beta3 and beta7 is over our threshold and is assumed to be divergent. In our third chain beta6 is approaching the threshold of concern.

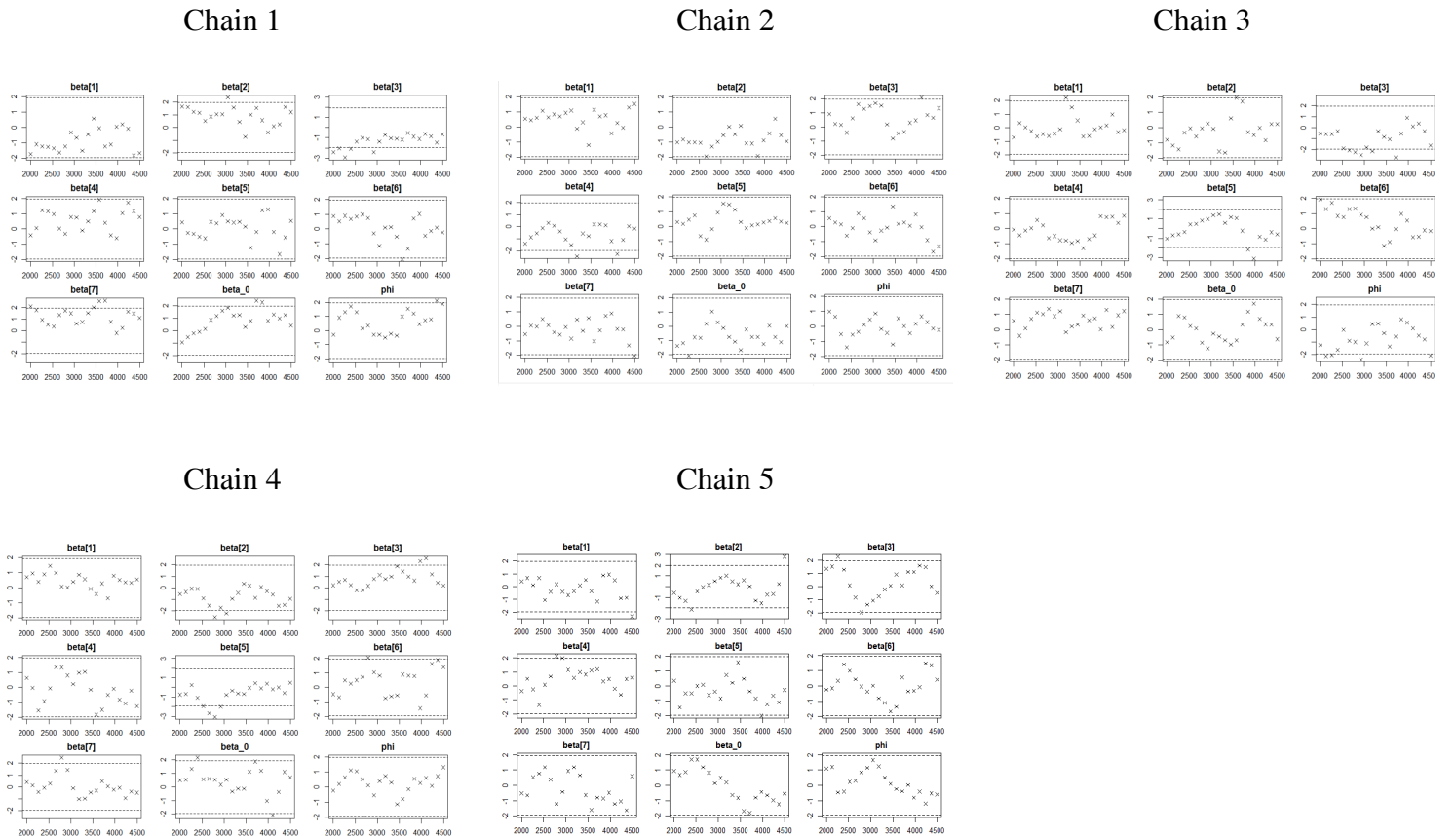


Figure 2. Geweke plot

Potential scale reduction factors:

	Point est.	Upper C.I.
beta[1]	1	1
beta[2]	1	1
beta[3]	1	1
beta[4]	1	1
beta[5]	1	1
beta[6]	1	1
beta[7]	1	1
beta_0	1	1
phi	1	1

Multivariate psrf

1

Table 3. Results of our Gelman-Rubin Diagnostics

According to our Gelman-Rubin Diagnostics it is seen that all point estimates for each of our parameters has a value of one meaning that the MCMC chains have converged and there is not a significant difference between within-chain variance and between-chain variance. The Upper Confidence Interval are all valued at one as well confirming convergence across all chains for each parameter.

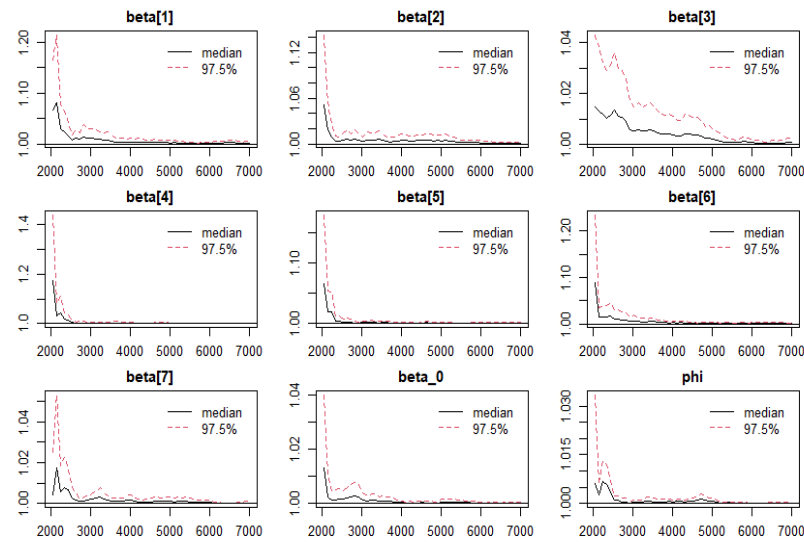


Figure 3. Gelman-Rubin plot for each parameter

Chain 1

	Burn-in (M)	Total (N)	Lower bound (Nmin)	Dependence factor (I)
beta[1]	15	18558	3746	4.95
beta[2]	14	15042	3746	4.02
beta[3]	7	7819	3746	2.09
beta[4]	7	8197	3746	2.19
beta[5]	7	7675	3746	2.05
beta[6]	14	14182	3746	3.79
beta[7]	6	6406	3746	1.71

Chain 2

	Burn-in (M)	Total (N)	Lower bound (Nmin)	Dependence factor (I)
beta[1]	15	17133	3746	4.57
beta[2]	12	14950	3746	3.99
beta[3]	12	14096	3746	3.76
beta[4]	9	9307	3746	2.48
beta[5]	7	7819	3746	2.09
beta[6]	14	15054	3746	4.02
beta[7]	7	7262	3746	1.94

Chain 3

	Burn-in (M)	Total (N)	Lower bound (Nmin)	Dependence factor (I)
beta[1]	14	14470	3746	3.86
beta[2]	16	16564	3746	4.42
beta[3]	7	8119	3746	2.17
beta[4]	7	7893	3746	2.11
beta[5]	6	6939	3746	1.85
beta[6]	14	16312	3746	4.35
beta[7]	6	6294	3746	1.68

Chain 4

	Burn-in (M)	Total (N)	Lower bound (Nmin)	Dependence factor (I)
beta[1]	12	12556	3746	3.35
beta[2]	14	15296	3746	4.08
beta[3]	12	12786	3746	3.41
beta[4]	9	9691	3746	2.59
beta[5]	7	7465	3746	1.99
beta[6]	16	17850	3746	4.77
beta[7]	6	6520	3746	1.74

Chain 5

	Burn-in (M)	Total (N)	Lower bound (Nmin)	Dependence factor (I)
beta[1]	12	13244	3746	3.54
beta[2]	12	14306	3746	3.82
beta[3]	7	8197	3746	2.19
beta[4]	7	7819	3746	2.09
beta[5]	7	7893	3746	2.11
beta[6]	12	14908	3746	3.98
beta[7]	6	6695	3746	1.79

Table 4. Raftery-Lewis Diagnostics

```
## Effective Sample Size
```{r}
effectiveSize(samples)
```
```

| | | | | | | | |
|-----------|----------|----------|----------|----------|----------|----------|-----------|
| beta[1] | beta[2] | beta[3] | beta[4] | beta[5] | beta[6] | beta[7] | beta_0 |
| 2652.491 | 2898.796 | 5370.582 | 4557.030 | 6138.848 | 2751.610 | 8513.012 | 13772.620 |
| phi | | | | | | | |
| 14622.371 | | | | | | | |

Figure 4. EffectiveSize Function Results

The ESS results reveal some autocorrelation in the sampling process, with relatively low values compared to the total iterations (Figure 4). However, convergence to independence was achieved at lag 0 for most chains, except chain 2 (lag 22).

```
Autocorr is 0 for chain 1 at lag -18
Autocorr is 0 for chain 2 at lag 22
Autocorr is 0 for chain 3 at lag -29
Autocorr is 0 for chain 4 at lag -13
Autocorr is 0 for chain 5 at lag -18
```

Table 5. Autocorrelation Result

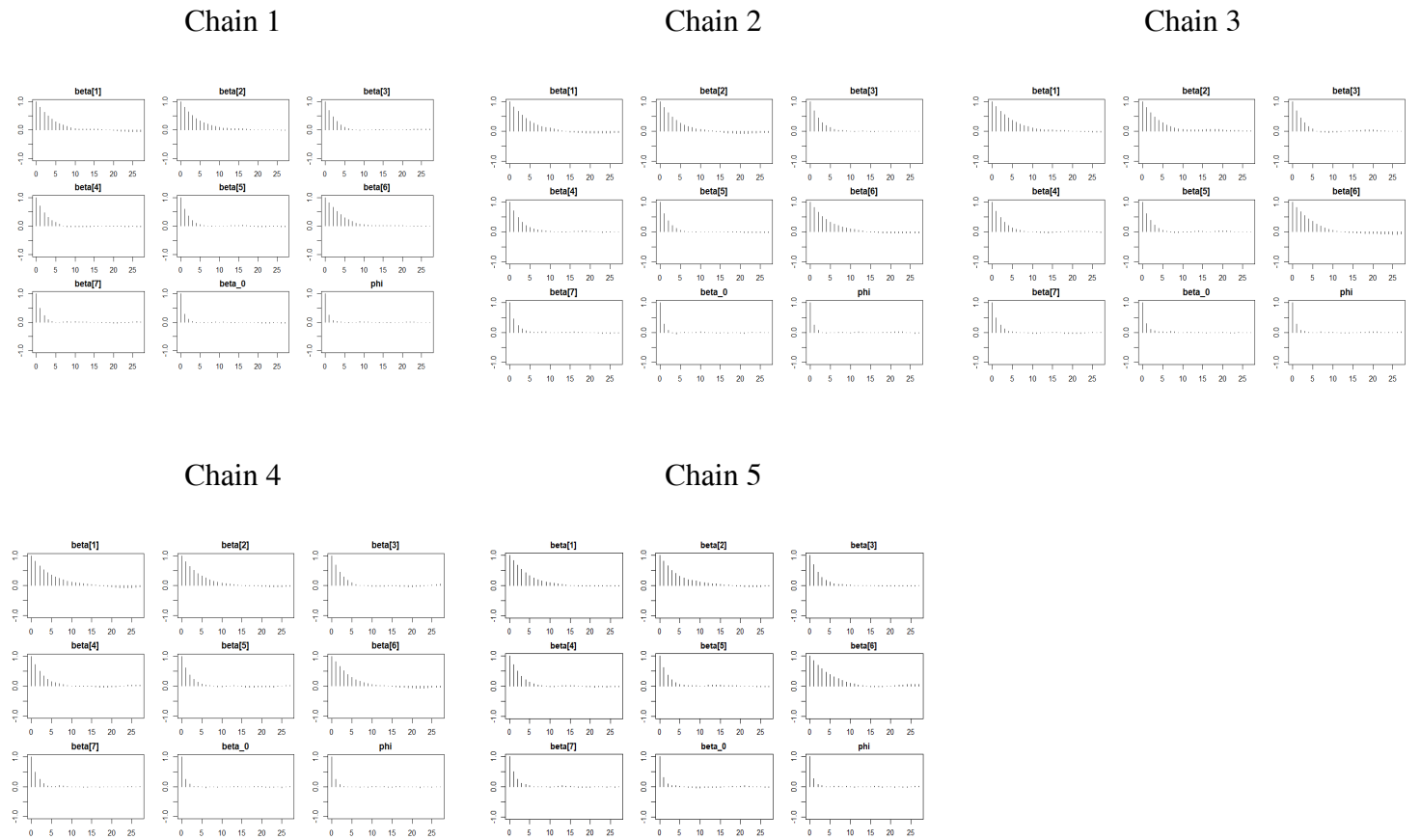


Figure 5: Autocorrelation Plot

The autocorrelation results indicate that all 5 chains eventually converge to 0, meaning that the sampled values are independent at specific lags. For chains 1,3,4, and 5, autocorrelation is already achieved at lag 1, considering that the value for those chains are less than 0. However, for chain 2, the chain eventually reached an autocorrelation at lag 22 (Figure 5).

Model Comparison

- WAIC (Watanabe-Akaike Information Criterion)

Comparison between the Beta regression model and itself highlights WAIC (Watanabe-Akaike Information Criterion) shows a score of -1118.597, with an effective complexity penalty (p-WAIC) of 9.435902. WAIC evaluates predictive accuracy while penalizing model complexity, and more negative values indicate a better balance between fit and overfitting. The effective complexity penalty, (p-WAIC) quantifies the number of parameters effectively contributing to the model's complexity. A value of 9.435902 suggests the model has moderate complexity, balancing flexibility and generalizability.

- LOO-CV (Leave-One-Out Cross-Validation)

| | Estimate
<S3: AsIs> | SE
<S3: AsIs> |
|----------|------------------------|------------------|
| elpd_loo | 708.8 | 20.9 |
| p_loo | 11.1 | 1.1 |
| looic | -1417.6 | 41.9 |

3 rows

Table 6. LOO-CV Results

The table summarized key metrics derived from LOO-CV (Leave-One-Out Cross-Validation), which is a robust technique for evaluating the predictive performance of Bayesian models. The Expected Log Predictive Density (elpd_loo) is 708.8 with a Standard Error (SE) of 20.9, reflecting the model's predictive accuracy, where higher values indicate better performance. The Effective Number of Parameters (p_loo), which measures model complexity, is 11.1 with an SE of 1.1, indicating the effective degrees of freedom used by the model while accounting for potential overfitting. Lastly, the LOO Information Criterion (LOOIC), which balances model fit and complexity, has a value of -1417.6 with an SE of 41.9, where lower values signify better model performance.

Posterior Predictive Checks

Graphical Representation

To aid with model evaluation, various graphs are provided in order to generally grasp the effectiveness of the model. To help with this process, a data frame containing the true values, predicted values, and residual values (obtained from the difference of the true and predicted values) is created.

Description: df [100 × 3]

| True
<dbl> | Predicted
<dbl> | Residual
<dbl> |
|---------------|--------------------|-------------------|
| 0.76 | 0.8003547 | -0.0403546583 |
| 0.75 | 0.7129708 | 0.0370292117 |
| 0.68 | 0.5599991 | 0.1200008793 |
| 0.84 | 0.8302064 | 0.0097935941 |
| 0.78 | 0.8428679 | -0.0628679250 |
| 0.65 | 0.6506469 | -0.0006469102 |
| 0.94 | 0.8900826 | 0.0499174298 |
| 0.97 | 0.9107017 | 0.0592983444 |
| 0.76 | 0.7674493 | -0.0074493221 |
| 0.48 | 0.5546542 | -0.0746542271 |

Table 7: Prediction Data Frame

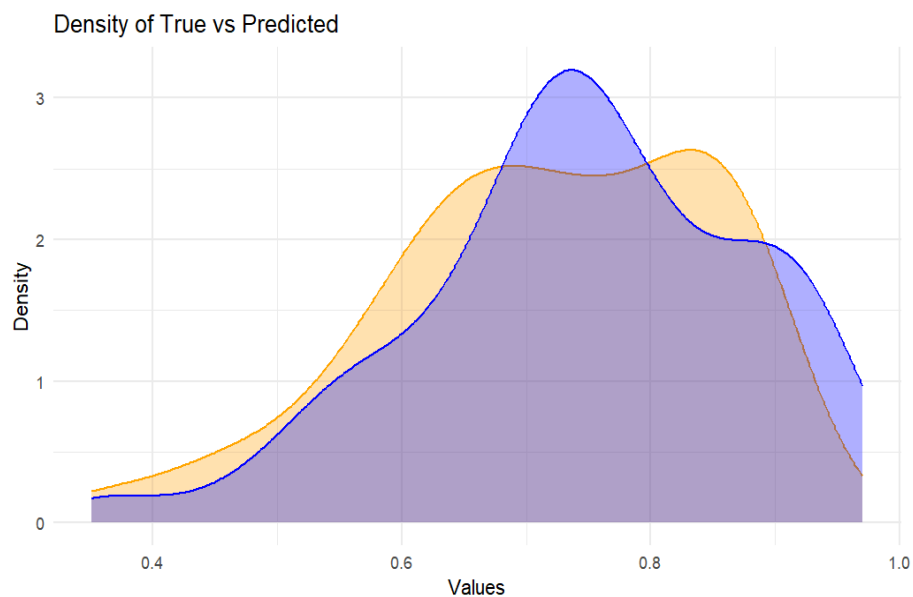


Figure 6 : Distribution of Predicted vs True Values' Density

This plot compares the density distributions of true values (in blue) and predicted values (in orange). The overlap between the two distributions indicates the degree to which the predictions match the true values. From the graph, it can be inferred that the predicted density closely aligns with the true density in some regions, especially near the peak values, suggesting the model captures the underlying data distribution well. However, there are slight deviations in areas where the distributions do not overlap perfectly, indicating room for improvement in the model's predictions.

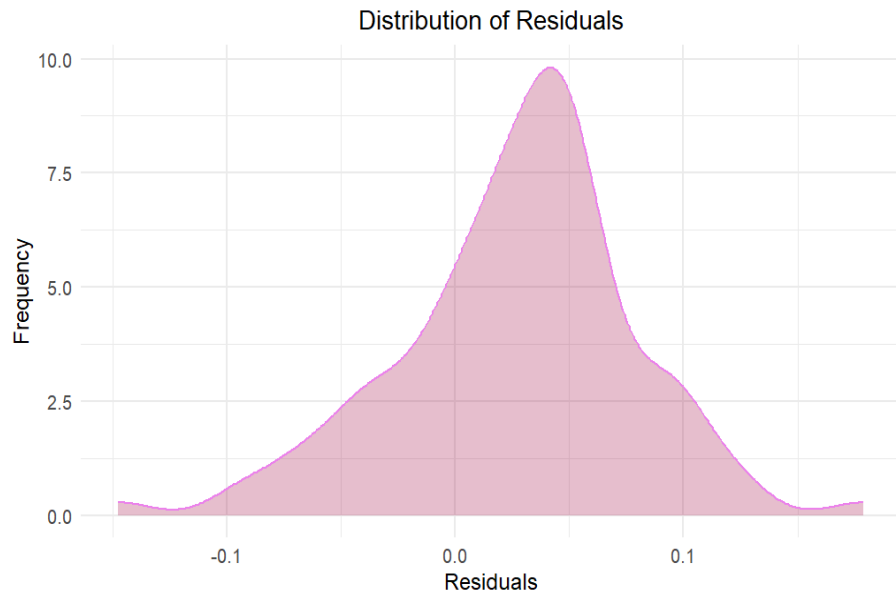


Figure 7 : Residual Distribution

The graph above showcases the density distributions of the residual values, which is obtained from calculating the difference between the true and predicted values. The graph shows that the residual values are centered and hardly deviates from 0.0 to 0.1, indicating that the error for the predicted values is not very significant.

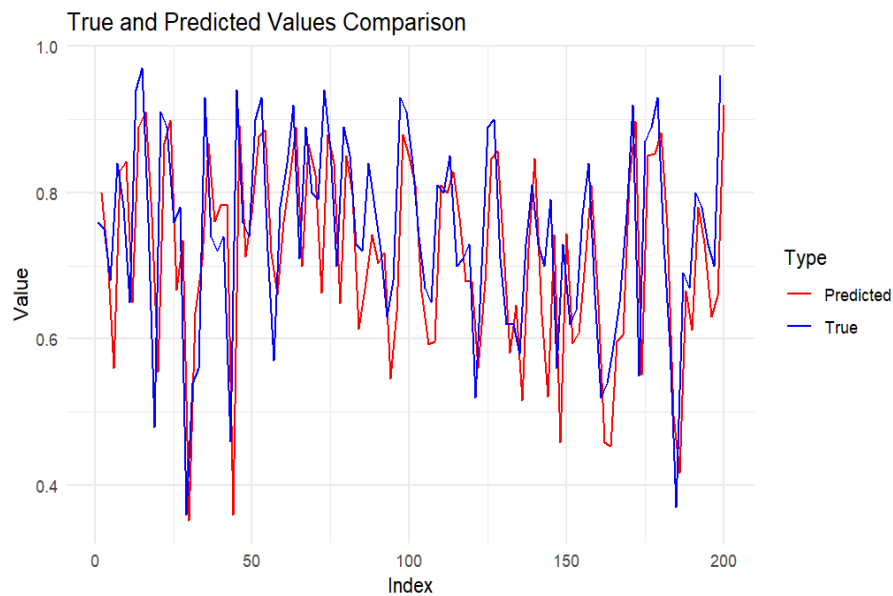


Figure 8 : Comparison Between Actual and Predicted Values by Index

The graph above shows the line graph for the true (colored blue) and predicted values (colored red) by index. Visually speaking, the graph doesn't seem to stray very far from each other, which implies that the predictions made are sufficiently accurate.

Statistical Metrics

Mean Squared Error (MSE)

The baseline MSE is recorded at 0.0179975, which represents the error of a simpler, initial model that serves as a benchmark. This value provides a reference point to measure the improvement of the trained model., whereas the trained model's MSE is reported as 0.003406591, showing a substantial decrease in error compared to the baseline. This result demonstrates the trained model's ability to predict with much greater accuracy.

Posterior-Predictive p-value (ppp-value)

A posterior predictive p-value indicates the model's ability to fit the data, and ranges from 0 to 1. The posterior predictive p-value – PPP-value for short, is calculated by taking the probability of the simulated data exceeding the actual data. From the model's simulation, a PPP-value of 0.45 is achieved.

DISCUSSION

Interpretation of The Results

Summary

The summary of the samples reveals some important details (refer to Table 1) – most notably, the empirical mean and standard deviation for each extracted parameter in the model, namely the intercept of the model (β_0), the slopes ($\beta_1, \beta_2, \beta_3, \dots, \beta_7$), and the precision parameter (Φ). Not only that, the summary also provides the 95% confidence interval for each aforementioned parameter, which is used to determine the overall significance of a column.

The significance of a column is achieved through verifying if the confidence interval of the corresponding slope value (e.g., β_1 is the slope for the first column – GRE.Score) passes through 0. This unconventional method is done in Bayesian Data Analysis because unlike frequentist confidence intervals, Bayesian credibility intervals offer a probabilistic interpretation: there is a 95% probability that the true value of a parameter lies within the specified interval, given the data and the prior. This approach aligns with the Bayesian philosophy of combining prior beliefs with observed data to update our understanding of the model parameters.

Given this knowledge we can infer that the third (University.Rating) and fourth column (SOP) has little significance in affecting the response variable outcome. Though Statement of Purpose letters and University ratings do contribute to the overall chance of admit, the scaling that was done before the modeling makes it so that the significance of these columns are overshadowed by much more meaningful covariates. For example, from Figure 1, we can also implicate the strong significance of CGPA (Cumulative Grade Point Average), with the corresponding slope (β_6) having a mean of 0.37348 when scaled.

The significance or insignificance of various columns can be validated by factoring in real-world college applications. For example, for university ratings, admission committees may give some weight to the reputation of an applicant's undergraduate institution, but this factor often serves as a tie-breaker rather than a primary criterion. Furthermore, the variability in ranking systems and biases can reduce its predictive power. As for Statement of Purpose, it provides qualitative insights into an applicant's goals, motivation, and alignment with the program.

However, its subjective nature and the potential for external assistance (e.g., professional editing) may diminish its direct impact in a quantitative model. On the other hand, CGPA is a direct measure of academic performance and consistency, often weighed heavily by admission committees. Its relatively high posterior mean suggests a strong and positive relationship with the outcome, reflecting the likelihood that higher CGPAs increase the probability of admission. This aligns well with real-world practices, where a strong academic record is one of the most critical factors in graduate admissions.

Trace and Density Plots

The plotting of the Markov Chain Monte Carlo chains display a set of trace plots and density plots. Trace plots are an essential diagnostic tool in Bayesian analysis as they provide a visual representation of the sampling process for each parameter. In the context of this model, the trace plots for each parameter—including the intercept (β_0), slopes ($\beta_1, \beta_2, \beta_3, \dots, \beta_7$), and the precision parameter (ϕ)—were analyzed to assess convergence and the quality of the Markov Chain Monte Carlo (MCMC) sampling.

For each parameter, the trace plots show a dense “hairy caterpillar” pattern, indicating that the chains mix well and sample effectively. This suggests that the sampler has reached a stationary distribution, meaning that the chains have similar means throughout all iterations (refer to Figure 1). The trace plots exhibit no discernible trends

or drifts over iterations, confirming that the chains have achieved stationarity, since any noticeable upward or downward trends would have indicated a lack of convergence.

Density plots on the other hand display the overall distribution of the beta (β) values based on the aforementioned confidence intervals. They aid in visualizing the data distribution of each included variable (refer to the Summary subsection for the conclusion).

Convergence Diagnostics

To better evaluate the convergence of the chains' samples, rather than just depending on visual means such as trace plots, JAGS has evaluation metrics to measure convergence for Bayesian modelling.

Geweke Diagnostics

Geweke Diagnostics is a type of convergence diagnostic, tailored to test the convergence of a singular Markov Chain Monte Carlo (MCMC) chain. It uses a statistical approach to compare the means between two different segments of the same chain, typically one from the beginning of the chain and the other at the end. The JAGS default for the segment size is the first 10% and the last 50% of the chain respectively. By analyzing the difference in these means relative to their variance, Geweke Diagnostics provides a standardized Z-score, where values far from zero may indicate that the chain has not yet converged to its stationary distribution.

The Z-score has a general formula of:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{Var(x_1) + Var(x_2)}}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the means of the first and last segments of the chain, and $Var(x_1)$ and $Var(x_2)$ are their respective variances. This formula quantifies the consistency between the early and late segments of the Markov Chain Monte Carlo (MCMC) samples. In the context of Geweke Diagnostics, a well-converged chain should have no significant difference between these two segments, resulting in a Z-score close to zero. A Z-score far from zero on the other hand suggests that the two segments differ significantly, implying that the chain may not yet have converged. This could necessitate further burn-in, longer chains, or reevaluation of the model. Generally speaking, the Z-score for the Geweke Diagnostic is compared to the Z-score of the corresponding confidence interval (± 1.96

for 95% confidence interval). Any less or more than said boundary can be interpreted as a bad Z-score, indicating that the chain does not converge.

JAGS provides built-in Geweke Diagnostics using the `geweke.diag` and `geweke.plot` syntax, where the former contains the Z-scores for each chain's parameters, and the latter function complements this by creating plots of the Z-scores across the chains. Each parameter's Z-score is plotted, often against the parameter index or another relevant measure, to provide a visual summary of how the Z-scores behave. Parameters with Z-scores consistently near zero across chains suggest proper convergence, while outliers or extreme values may signal a lack of convergence or mixing problems. Using these tools together provides a comprehensive approach to diagnosing chain convergence in Bayesian models.

Referring to Figure 2, we can see that a sizable amount of coefficients in Chain 1 exceed the aforementioned boundaries, specifically for β_3 and β_7 . This indicates that the values for the corresponding slopes (University.Rating for β_3 and Research for β_7) did not converge, thus making the chain unusable for reliable inference.

The plots from Figure 6-10 shows us the z-score of each chain throughout the 5.000 iterations, indicated by the crosses (x) in the plots. For most parameters, the Z-scores are distributed near zero, suggesting convergence to the stationary distribution. However, for β_3 (University Rating) and β_7 (Research), Chain 1 exhibits Z-scores significantly outside the ± 1.96 confidence interval, highlighting a lack of convergence. This behavior is not observed uniformly across all chains, indicating that the convergence issues may stem from poor mixing or specific initialization conditions in Chain 1. Most parameters maintain Z-scores close to zero, suggesting convergence. However, the notable divergence in Chain 1 for and highlights poor mixing or insufficient burn-in. Addressing this issue could involve extending the chain length, refining the priors for these parameters, or diagnosing potential model misspecifications.

Gelman-Rubin Diagnostics

The Gelman-Rubin Diagnostic, also known as the Potential Scale Reduction Factor (PSRF), is a widely used metric to assess the convergence of multiple Markov Chain Monte Carlo (MCMC) chains. Unlike Geweke Diagnostics, which analyzes a single chain, the Gelman-Rubin Diagnostic compares the variability within each

chain to the variability between chains. The goal is to evaluate whether all chains are sampling from the same stationary distribution.

The Gelman-Rubin diagnostic is defined as:

$$\hat{R} = \frac{\hat{V}}{W}$$

where $\hat{V}$ is defined as the pooled variance estimate and W is defined as the within-chain variance estimate. The formula for these parameters won't be explained in-depth, and it is encouraged to visit the referenced articles for more clarity.

$\hat{R}$ (the Potential Scale Reduction Factor) is then compared to 1 to assess the convergence of the chains. When $\hat{R}$ (the Potential Scale Reduction Factor) is close to 1, it suggests that the chains have likely converged and are sampling from the same stationary distribution. Specifically:

- $\hat{R} = 1$ indicates perfect convergence as the variability within each chain (W) is similar to the variability across chains ($\hat{V}$).
- $\hat{R}$ between 1 and 1.1 implies adequate convergence, with the variability across chains to be slightly higher than the variability within each chain.
- $\hat{R}$ above 1.1 implies divergence, with higher values indicating greater discrepancy between within-chain and between-chain variances.

Referring to the Gelman-Rubin diagnostic simulated in JAGS, we can observe that the PSRF (Potential Scale Reduction Factor) in Figure 3 is consistently equal to 1 across all monitored parameters. This result strongly indicates that the chains have achieved perfect convergence, with no significant differences between the within-chain and between-chain variances. Such a value reflects that the chains are effectively sampling from the same stationary distribution, confirming the reliability of the posterior estimates.

This outcome underscores the robustness of the model setup, as the initial values for the chains appear to lead to consistent results, regardless of their starting points. Achieving a PSRF of 1 across all parameters validates the sufficiency of the burn-in period and the number of iterations used in the simulation. It also suggests that the

parameters of interest are well-identified and that the sampling process was efficient. These results provide a solid foundation for making reliable inferences about the model's parameters.

Raftery-Lewis Diagnostics

The Raftery-Lewis Diagnostics is a Bayesian evaluation tool provided in JAGS designed to assess the adequacy of a Markov Chain Monte Carlo (MCMC) sample for Bayesian inference. Specifically, it evaluates the number of iterations required for the MCMC chain to estimate a specified quantile of the posterior distribution with a desired level of precision and a set probability. The purpose of the Raftery-Lewis Diagnostic is to determine:

- The sufficient amount of iterations needed to estimate a given quantile with a specific precision (represented by n)
- The burn-in period, which accounts for the early iterations of the chain that may not yet represent the stationary distribution (represented by M), and
- The degree of autocorrelation present in the Markov Chain Monte Carlo (MCMC) chain, which affects the efficiency of the sampling process (represented by I).

Generally, the Raftery-Lewis Diagnostic is used in fine-tuning the model, such as finding out the optimal burn-in period and iterations. However, for this specific case the focus lies in the Dependence Factor (I) in order to find the degree of autocorrelation for each chain, where:

- $I \approx 1$ suggests little to no autocorrelation, indicating that the chain is sampling independently.
- I between 2 and 4 suggests moderate autocorrelation. While the chain can still produce reliable results, efficiency is reduced, which may need tuning, be it making longer chains or thinning to ensure sufficient independent samples.
- $I > 5$ suggests significant autocorrelation. This can lead to poor mixing, and the chain may not provide reliable results within a reasonable number of iterations. It's a sign to investigate further, such as increasing the iteration or improving the sampling algorithm.

Based on the Raftery-Lewis Diagnostic results for this analysis (refer to Table 4), the Dependence Factors (I) for the majority of parameters fall within the range of 2 to 4. This range indicates that the chains exhibit

moderate autocorrelation, meaning successive samples are somewhat correlated but not excessively so. Moderate autocorrelation suggests that while the chains may not be fully efficient in producing independent samples, they are still capable of providing reliable results, provided the total number of iterations is sufficient.

The diagnostic also highlights the varying levels of sampling efficiency across parameters. For instance, the Dependence Factor (I) of β_1 is closer to the upper limit in most cases, suggesting reduced efficiency compared to other parameters like β_7 , which has Dependence Factors that are much closer to 1, indicating that its samples are more independent and that the sampling process for this parameter is more efficient. Despite these variations, the results remain within acceptable bounds for effective inference.

Effective Sample Size

The Effective Sample Size (ESS) is a metric used in MCMC analysis to quantify the amount of independent information present in a set of correlated samples. While MCMC produces many iterations, the samples are often autocorrelated due to the nature of the algorithm. ESS estimates the number of independent (uncorrelated) samples that are equivalent to the information contained in the correlated chain. A higher ESS indicates better sampling efficiency and more reliable parameter estimates.

Generally speaking, ESS doesn't have a set threshold, unlike Geweke and Gelman-Rubin diagnostics. ESS results are heavily dependent on the number of iterations the MCMC sample has. A higher ESS is typically desirable, as it indicates that the MCMC chains have explored the parameter space effectively, producing more independent and reliable samples. A low ESS suggests high autocorrelation between samples, which reduces the effective information content of the chains and may signal the need for adjustments to the sampling process.

From the presented results (Figure 4), some autocorrelation can be revealed from the results. Some parameters, like β_0 and ϕ have a comparatively large effective sample size. These variables specifically have little to no autocorrelation. Take the intercept (β_0) for instance, which is typically less affected by individual covariates like β_1 , or β_2 because it represents the baseline of the response variable when all predictors are zero. As for the precision parameter (ϕ), it is responsible for capturing the variability or dispersion in the response variable. Since it operates independently of the predictors' effects, its posterior distribution may be less complex and not as correlated towards the other variables.

Slopes, on the other hand tend to exhibit stronger autocorrelation because they are influenced by multicollinearity or shared/redundant information between predictors. For instance, if some predictors are correlated, the Markov Chain Monte Carlo (MCMC) sampler may have difficulties moving efficiently through the posterior space for the corresponding slope parameters.

Autocorrelation

Autocorrelation analysis is a critical step in validating the convergence and efficiency of Markov Chain Monte Carlo (MCMC) simulations, especially in Bayesian analysis. In this study, the autocorrelation results for the five chains, along with the associated plots for parameters $\beta[1]$ through $\beta[7]$, β_0 , and ϕ , provide insight into the mixing and independence of the samples drawn.

The reported autocorrelation values indicate that all chains reach an autocorrelation of zero at varying lags:

- Chain 1: Lag -18
- Chain 2: Lag 22
- Chain 3: Lag -29
- Chain 4: Lag -13
- Chain 5: Lag -18

These results suggest that the chains exhibit different levels of dependence between successive samples, with Chain 3 achieving decorrelation at a slightly higher lag compared to the others.

Implications for Convergence:

The consistent behavior of all chains, despite minor variations in lag thresholds, supports the conclusion that the MCMC simulation has converged. The absence of persistent high autocorrelation values reinforces the validity of the posterior estimates derived from these simulations.

In conclusion, the autocorrelation results demonstrate that the MCMC chains employed in this study exhibit satisfactory mixing and independence properties. These findings enhance confidence in the robustness of the Bayesian inference conducted and provide a reliable foundation for subsequent analysis.

Posterior Predictive Checks

Posterior Predictive Checks (PPCs) are the key diagnostic method in Bayesian statistics used to evaluate how effectively a model represents the observed data. The process involves generating new data samples from posterior predictive distribution, which is expressed as:

$$p(\tilde{y} | y) = \int p(\tilde{y} | \theta) p(\theta | y) d\theta$$

Where θ denotes the model parameters, $p(\tilde{y} | \theta)$ represents the likelihood of the simulated data given the parameters, and $p(\theta | y)$ corresponds the posterior distribution of the parameters given the observed data y . PPCs are essential for identifying model misspecifications, ensuring the model captures key patterns in the data and providing insights into uncertainty and predictive performance, thereby guiding effective model refinement.

In order to simulate prediction in Bayesian Data Analysis, samples from the model parameters (θ) is generated from their posterior distribution and uses these samples to generate new data points. These new data points will be affected by the covariates from the test dataset that was previously declared, and then compared to the actual/true value.

In this study, the simulated data will be calculated using the **rbeta** syntax, which simulates data points following the beta distribution. This is particularly relevant for Bayesian beta regression, where the response variable is modeled using a beta distribution constrained between 0 and 1. The **rbeta** function uses the shape parameters (α and β), which are derived from the posterior samples of the regression model.

The predictive distribution is obtained by incorporating the covariate effects into the posterior samples of the model parameters. For example, the shape parameters of the beta distribution may be calculated using the model's linear predictor transformed through a link function. The simulated data generated from these distributions can then be compared to the observed data to assess how well the model captures the patterns and variability in the data.

These predictive values' mean are then calculated to provide a summary measure of the central tendency of the predicted outcomes. This allows for a direct comparison to the observed data's mean, offering insights into whether the model systematically overestimates or underestimates the observed values.

Mean Squared Error (MSE) serves as a quantitative measure of the average squared differences between the observed values (y) and the predicted values ($\tilde{y}$). Calculated as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2$$

A lower MSE indicates a better fit, as it demonstrates that the predicted values are closer to the observed data. Posterior predictive p-value provides a diagnostic measure to assess whether the model-generated data is consistent with the observed data. Calculated by comparing a chosen test statistic such as mean or variance, between observed and simulated data:

$$p_{post} = P(T(\tilde{y}) \geq T(y) | y)$$

Values close to 0.5 indicate a good match between the simulated and observed data, while extreme values (near 0 and 1) signal potential model misspecifications. For this case, the PPP-value for the model is **0.45**. This indicates the excellent fit of the model, where the model doesn't overfit or underfit.

Comparison with Existing Literature

To better assess the model, the best course of action is to compare the results with existing models. The compared model is using a different method, using R2JAGS as the foundation instead of the usual RJAGS.

| Existing Literature | Proposed Model |
|---|---|
| Applied Bayesian Model Averaging | Due to us only using one model we did not apply Bayesian Model Averaging |
| Only used Mean Squared Error as the evaluation metric | Used Posterior-Predictive p-value (ppp-value) as well as Mean Squared Error (MSE) as evaluation metrics |
| A simpler model resulting in a higher MSE score showing the model's is less accurate compared to the proposed model | A more complex yet more resource-intensive model resulting in a lower MSE score (0.003406591) demonstrating a more accurately predicting model. |

Implications of the Findings

The findings of our project have several significant implications for predictive modeling in the context of graduate admissions.

The proposed model, which utilizes both Mean Squared Error (MSE) and Posterior Predictive p-values (ppp-values) as evaluation metrics, demonstrates superior predictive accuracy compared to simpler models. The significantly lower MSE score (0.003406591) highlights its ability to more effectively predict admission probabilities, providing a robust tool for decision-making in academic admissions processes.

The resource-intensive nature of the proposed model underscores the trade-off between computational complexity and predictive accuracy. While the model requires greater computational resources, its enhanced accuracy justifies the cost in contexts where precise predictions are critical, such as competitive graduate admissions.

The comparison with Bayesian Model Averaging (BMA) from the existing literature reveals that simplicity does not always equate to optimality. The higher MSE score of the simpler BMA approach suggests that relying solely on simpler models or a single evaluation metric could lead to suboptimal decision-making in high-stakes scenarios.

CONCLUSION

Summary of the Study

This study evaluates the performance and convergence of Bayesian Beta Regression model for predicting academic admission probabilities. The model demonstrated successful convergence across Markov Chain Monte Carlo (MCMC) chains, with the Gelman-Rubin and Geweke diagnostic confirming appropriate convergence of all parameters. Despite some autocorrelation observed in the chains, the results confirmed satisfactory mixing and independence, ensuring the reliability of the inference. Evaluation metrics, including Watanabe Akaike Information Criterion (WAIC), Leave-Out-One Cross-Validation (LOO-CV), and Mean Squared Error (MSE), indicated that the model fits the data well without overfitting. Posterior predictive checks further reinforced the model's predictive accuracy, with a low MSE and a posterior predictive p-value (PPP), signaling a good match between simulated and observed data. A comparison with existing literature highlighted the superiority of the proposed model in terms of

predictive performance, showcasing its potential for improving decision-making in academic admission. Overall, the study's results suggest that the Bayesian model provides an accurate and reliable framework for assessing complex academic decision-making processes.

Purpose and Context

This study aims to properly evaluate a student's chance of admission to a selected university by using Bayesian methods. Among these methods, Bayesian Beta Regression is utilized to model the probability of admission, as it is particularly suited for analyzing outcomes that fall within a bounded range, such as probabilities or proportions. By incorporating prior knowledge and updating beliefs with observed data, the Bayesian approach allows for more nuanced insights compared to traditional regression techniques. This methodology not only accounts for the inherent uncertainty in admission predictions but also provides a flexible framework for incorporating additional contextual factors, such as a student's academic achievements, extracurricular involvement, and demographic background. The goal is to produce a robust, interpretable model that can better capture the complexities of university admission processes and offer practical implications for students, educators, and policy-makers alike.

Key Findings

The analysis successfully demonstrated the applicability of Bayesian Beta Regression in modeling admission probabilities for university applicants. The Bayesian Beta Regression model demonstrated robust predictive accuracy, as evidenced by the Mean Squared Error (MSE) of **0.003406591** and flexibility in handling bounded outcomes, as shown by the Posterior Predictive p-value (PPP-value) of **0.45**. Compared to frequentist regression techniques, it accounted more effectively for variability and uncertainty in the data.

This study also revealed some key correlations between the **chance of admission** and other variables, like how a strong positive relationship between the Cumulative GPA (CGPA) and admission probabilities was revealed, suggesting that academic performance is a key determinant in the selection process. Furthermore, factors such as extracurricular activities and personal achievements were also found to significantly enhance the likelihood of admission.

The results underscore the value of Bayesian Beta Regression in applications involving bounded outcomes, such as admission probabilities. For students, the findings highlight the importance of excelling in academic metrics and showcasing diverse achievements to improve their chances of acceptance. For universities, the model provides a data-driven approach to refine selection criteria and enhance decision-making fairness.

Overall, the findings illustrate the utility of Bayesian Beta Regression as a robust tool for modeling complex, real-world decisions. The low MSE further supports its effectiveness and reliability in predicting bounded probabilities, making it a promising approach for similar applications in educational analytics and beyond.

Limitations

While this study highlights the effectiveness of Bayesian Beta Regression in modeling proportions and bounded continuous outcomes, several limitations must be acknowledged. One of the key limitations of this study lies in the dataset used for developing and evaluating the Bayesian Beta Regression model. The dataset's scope and quality play a crucial role in determining the comprehensiveness and accuracy of the model. In this case, the data used may not fully capture the variability and complexity present in broader real-world scenarios. Limitations such as a small sample size, unbalanced representation, or potential data imperfections, including noise or missing values, may have introduced biases into the analysis. Consequently, while the findings are relevant within the specific context of this dataset, their generalizability to other settings remains uncertain. Addressing this limitation in future studies by employing larger, more diverse datasets would enhance the robustness and applicability of the model's conclusions.

Another significant limitation stems from the incomplete exploration of model comparison. Due to the lack of adequate references and the challenges associated with deprecated or outdated documentation for alternative methodologies, this study did not fully benchmark the Bayesian Beta Regression model against other approaches. A comprehensive comparison is essential to evaluate the relative strengths and weaknesses of different models and to situate the findings within the broader context of existing statistical methods. Future research could address this gap by conducting a systematic comparison using well-documented and validated methodologies, thereby offering a clearer perspective on the advantages and potential trade-offs of Bayesian Beta Regression relative to other techniques.

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